

Problem

Design a problem to help other students better understand how to find the initial and final value of a transfer function.

Step-by-step solution

Step 1 of 2

Consider a function

$$H(s) = \frac{a_1 s^{n-1} + a_2 s^{n-3} + a_3 s^{n-5} + \dots + a_4}{b_1 s^m + b_2 s^{m-2} + b_3 s^{m-4} + \dots + b_4}$$

Thus, using initial value theorem

$$h(0) = \lim_{s \rightarrow \infty} sH(s)$$

$$h(0) = \lim_{s \rightarrow \infty} s \left[\frac{a_1 s^{n-1} + a_2 s^{n-3} + a_3 s^{n-5} + \dots + a_4}{b_1 s^m + b_2 s^{m-2} + b_3 s^{m-4} + \dots + b_4} \right]$$

$$h(0) = \lim_{s \rightarrow \infty} \frac{s^n \left(a_1 + \frac{a_2}{s^2} + \frac{a_3}{s^4} + \dots + \frac{a_4}{s^{n-1}} \right)}{s^m \left(b_1 + \frac{b_2}{s^2} + \frac{b_3}{s^4} + \dots + \frac{b_4}{s^m} \right)}$$

When $m > n$, then

$$h(0) = \lim_{s \rightarrow \infty} \frac{\left(a_1 + \frac{a_2}{s^2} + \frac{a_3}{s^4} + \dots + \frac{a_4}{s^{n-1}} \right)}{s^{m-n} \left[b_1 + \frac{b_2}{s^2} + \frac{b_3}{s^4} + \dots + \frac{b_4}{s^m} \right]}$$

$$\therefore h(0) = 0$$

When $m < n$, then

$$h(0) = \lim_{s \rightarrow \infty} \frac{s^{n-m} \left[a_1 + \frac{a_2}{s^2} + \frac{a_3}{s^4} + \dots + \frac{a_4}{s^{n-1}} \right]}{\left[b_1 + \frac{b_2}{s^2} + \frac{b_3}{s^4} + \dots + \frac{b_4}{s^m} \right]}$$

$$\therefore h(0) = \infty$$

Step 2 of 2

When $m = n$, then

$$h(0) = \lim_{s \rightarrow \infty} \frac{a_1 + \frac{a_2}{s^2} + \frac{a_3}{s^3} + \dots + \frac{a_4}{s^{n-1}}}{b_1 + \frac{b_2}{s} + \frac{b_3}{s^4} + \dots + \frac{b_4}{s^m}}$$

$$\therefore h(0) = \frac{a_1}{b_1}$$

Therefore, using final – value theorem

$$h(\infty) = \lim_{s \rightarrow 0} sH(s)$$

$$h(\infty) = \lim_{s \rightarrow 0} s \left[\frac{a_1 s^{n-1} + a_2 s^{n-3} + a_3 s^{n-5} + \dots + a_4}{b_1 s^m + b_2 s^{m-2} + b_3 s^{m-4} + \dots + b_4} \right]$$

$$h(\infty) = \lim_{s \rightarrow 0} s \frac{a_1 s^n + a_2 s^{n-2} + a_3 s^{n-4} + \dots + a_4 s}{b_1 s^m + b_2 s^{m-2} + b_3 s^{m-4} + \dots + b_4}$$

$$\therefore h(\infty) = 0$$